ISA/IEC 62443-3-3 / 4-2 • SL-3 TESTING CHEAT SHEET

Moxa EDS / MDS Managed Industrial Ethernet Switch (EDS-4000/G4000, MDS-G4000) · Component type: **Network Device (NDR) - managed industrial Ethernet switch (IEC 62443-4-2)** · Target SL = SL 3

Where to work	The switch **web UI (HTTPS)** and **CLI (SSH)**; **System › Account Management** (users/roles, password policy, login lockout); **Security** menus (802.1X, ACL, DHCP snooping, storm control, MAC sticky); firmware/secure-boot status; managed at scale by **MXview / MXsecurity**. Default login **admin/moxa** (change it).
Ports / paths	**HTTPS 443** + **SSH 22** (management); **HTTP 80 / Telnet 23** (disable); **SNMP 161** (v1/v2c/v3 - use v3); **Syslog 514**; RADIUS 1812/1813; TACACS+ 49; NTP 123; 802.1X (EAPOL). Physical: RJ45/console, reset, DI/DO relay; DIN-rail. Change the default IP.
Firmware / platform	Moxa **EDS-4000/G4000** / **MDS-G4000** = **IEC 62443-4-2 certified** (secure boot + signed firmware + strong login policy). Older **EDS-405A/408A/G512** are lighter - confirm the model + firmware and note reduced capability. Track Moxa PSIRT advisories and apply firmware.
CLI syntax (Moxa — two OS generations)	Moxa CLI is Cisco-*like* but not identical, and splits by generation: **next-gen "Managed Switch OS" (v4.x, EDS-4000/G4000 & MDS-G4000)** vs **classic firmware (FW_5.x, EDS-510E/G-series, IKS)**. Both support `show running-config` and `show interfaces`, but menus/verbs differ (e.g. account/policy commands), so confirm against the matching Moxa CLI manual for the firmware. Older **EDS-405A/408A/G512** are largely **web-UI-only** with a minimal CLI — do the review in the web UI there. Verify firmware with `show version` / `show system` (web: System › About). SNMPv3 read-only is the safest evidence path across all models.
How to read	Default read-only (web/CLI read-only account, read-only SNMPv3). Each row = native step-by-step (web/CLI) + command-level external-tool validation + pass. Write/exploit tools = isolated env + written authorization. Modern Moxa is secure-by-design - most rows Met when hardened.

Session setup & evidence handling

Do this BEFORE running any command so the evidence is complete, attributable, and protected:
1. Start a full session log first. PuTTY: Session › Logging → select **All session output** → set **Log file name** (browse to the encrypted evidence folder) → under “if the log file already exists” pick **Always append** → tick **Include header** and (Logging › Options) **Flush log frequently**; put `&Y&M&D_&T` in the name for an auto timestamp. (SecureCRT: File › Log Session; Tera Term: File › Log; macOS/Linux: `script -a <file>.log` or `ssh user@host | tee <file>.log`.)
2. Name the file to the evidence convention (one file per device): `<site>_<zone>_<assetTag>_<vendor>_<model>_<mgmtIP>_<YYYYMMDD-HHMM>_<testerInitials>.log`.
3. Disable paging so full output is captured (it resets each login): disable paging (Managed Switch OS v4.x): `terminal length 0`; on lighter/older models with no reliable pager, use the web UI Export or read-only SNMPv3 instead.
4. Self-document the log - capture identity first: `show version` (web: System › About), so the file proves what/when/who.
5. Close out: restore paging, stop the log (PuTTY closes it when the session ends), and verify the file opened/saved.
6. Protect the evidence file. It contains running-config and may hold hashes, SNMP community strings, or keys: store **only** on the encrypted assessment share, follow the project data-classification/retention policy, redact secrets before sharing, and **never** email it in cleartext. **Do not save the device password in the PuTTY/SecureCRT saved session** - authenticate interactively or via the approved password manager.

FR 1 — Identification & Authentication Control (IAC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 1.1	Human user I&A	Confirm web (HTTPS) + CLI (SSH) require login and the **default admin/moxa is changed** (System › Account Management). Use named accounts + roles; integrate **RADIUS/TACACS+** for central auth where available. Disable HTTP/Telnet. Port-level device auth via **802.1X**. **Test:** browse HTTP/Telnet -> refused; HTTPS/SSH without creds -> denied; default admin/moxa rejected. **Ext. tools:** `nmap -p 22,23,80,443 <sw>` (only 22/443); RADIUS test; `ssh -v admin@<sw>`.	HTTPS/SSH authenticate; HTTP/Telnet off; default changed; RADIUS/TACACS+ where used.	Telnet+HTTP off for SL-3.

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 1.2	Device & software-process I&A	Port authentication via **802.1X** (Security › 802.1X) with RADIUS, plus **sticky MAC / MAC protections**; confirm switch-to-switch/uplink identity. SNMP requires v3 with auth. **Test:** a rogue endpoint on an 802.1X port lands in the guest/restricted VLAN or is denied. **Ext. tools:** 802.1X test supplicant + rogue device; Wireshark `eapol`; RADIUS logs.	Only authenticated devices connect; unknown endpoints quarantined; SNMPv3.	
SR/CR 1.3	Account management	In Account Management review named users/roles (admin/supervisor/user), least privilege, remove/rename defaults, no shared logins; where RADIUS/TACACS+ is used, manage centrally. **Ext. tools:** Web (Account Management export); RADIUS/TACACS+ server review.	Named least-privilege accounts; defaults removed; central auth where used.	
SR/CR 1.4	Identifier management	Confirm unique user identifiers and a unique device identity (hostname/IP - change the default, MAC/serial); no reuse. **Ext. tools:** Web/CLI device info; `nmap -sU -p 161 --script snmp-info`; inventory.	Unique user and device identifiers; no reuse.	
SR/CR 1.5	Authenticator management	In Account Management confirm the **password policy** (min length/complexity/expiry) and change all defaults; manage SSH keys / HTTPS certs. Strong RADIUS/TACACS+ shared secrets. **Ext. tools:** Web (password-policy screenshot); `ssh-audit <sw>`.	Strong password policy enforced; defaults changed; keys/certs managed.	
SR/NDR 1.6	Wireless access management	**N/A.** Wired switch with no native wireless interface. **Ext. tools:** Confirm no WLAN interface.	N/A - justification recorded (wired network device).	N/A
SR/CR 1.7	Strength of password-based authentication	Confirm the password-strength policy (length/complexity/lifetime) meets SL-3. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.5** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of password-based authentication verified individually.	
SR/CR 1.8	Public key infrastructure (PKI) certificates	Confirm the **HTTPS/SSH** certificates/keys are CA-signed/valid (import your cert; replace the default self-signed), strong (>=2048 RSA/ECC); 802.1X/RADIUS uses certs where configured. **Ext. tools:** `sslscan <sw>:443` / `testssl.sh https://<sw>`; `ssh-audit <sw>`.	CA-signed HTTPS cert; strong SSH keys/ciphers; defaults replaced.	
SR/CR 1.9	Strength of public key-based authentication	Confirm public-key authentication uses strong, current keys/algorithms. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.8** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of public key-based authentication verified individually.	
SR/CR 1.10	Authenticator feedback	Confirm the web/CLI login masks password entry (no cleartext echo) and stored credentials are hashed (no cleartext in the config export). **Ext. tools:** Visual verification; config export (no cleartext passwords).	Authenticators obscured during entry and in config.	
SR/CR 1.11	Unsuccessful login attempts	Confirm the **login lockout** policy (Account Management) - account locks after N failed attempts for a lockout period; failures are logged/trapped. **Test:** exceed the failed-login threshold (authorized) -> account locks. **Ext. tools:** Web (lockout policy); Hydra (ISOLATED, authorized) to confirm lockout engages.	Login lockout enabled and effective; failures logged.	
SR/CR 1.12	System use notification	Confirm a login banner/use-notification where supported (web/CLI login message); also enforce it on the management workstation (GPO). **Ext. tools:** Web/CLI banner config; EWS GPO.	Approved use notice at the switch login + EWS.	
SR/NDR 1.13	Access via untrusted networks	Restrict management: allowed-IP / management-interface access control (Security), SSH/HTTPS only, a dedicated management VLAN; remote admin via VPN + RADIUS/MFA; no management from user/untrusted VLANs. **Test:** from an untrusted VLAN, reach 22/443 -> must fail. **Ext. tools:** `nmap -Pn -p 22,443,161 <sw>` from each VLAN; management-access-control review.	Management-plane unreachable from untrusted networks; MFA/RADIUS for remote.	

FR 2 — Use Control (UC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 2.1	Authorization enforcement	Confirm role-based least privilege (admin / supervisor / user) - a user-level login cannot change configuration; where TACACS+ command authorization is available, use it. **Test:** as a user-level login, attempt a config change -> denied and logged. **Ext. tools:** Web (role/permission review); user-level negative test; TACACS+ logs.	Least privilege enforced per role; unauthorized actions denied and logged.	
SR/CR 2.2	Wireless Use Control	**N/A.** No wireless interface on this device; wireless access is managed at the network layer.	N/A - justification recorded (not assessed as a control).	N/A
SR 2.3	Use control for portable/mobile devices	Control the engineering laptop and any USB/config media; removable-media policy on the EWS; only authorized firmware/config loaded (signed). **Ext. tools:** EWS removable-media policy; firmware/config provenance.	Portable/engineering media controlled.	
SR/NDR 2.4	Mobile code	**N/A / limited.** No general-purpose app-hosting/mobile-code runtime; confirm only signed vendor firmware + the web UI run (no user script path). **Ext. tools:** Confirm no scripting/app-hosting; web content vendor read-only.	No mobile-code runtime; only signed firmware runs.	NDR-specific.
SR/CR 2.5	Session lock	Confirm web/CLI sessions time out on inactivity (re-login required); enforce the host OS lock on the management workstation. **Ext. tools:** Web/CLI timeout config; host inactivity-lock GPO.	Sessions time out on inactivity; re-auth required.	
SR/CR 2.6	Remote session termination	Confirm management sessions can be viewed/terminated and idle sessions close; concurrent management sessions are bounded (Account Management / session settings). **Ext. tools:** Web/CLI (active sessions); Wireshark (session teardown).	Sessions terminable; concurrency bounded/monitored.	
SR/CR 2.7	Concurrent session control	Confirm concurrent sessions are bounded/limited and monitored. **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Concurrent session control verified individually.	
SR/CR 2.8	Auditable events	Confirm the event log + **syslog** (local + remote to the SIEM) capture logins, config changes, 802.1X, and security events with the user identity; per-user accounts give attribution. **Test:** change config -> a syslog event with user + action + time reaches the SIEM. **Ext. tools:** Web (syslog config); `nc -lu 514` + SIEM; trigger a change.	Security events logged with attribution and forwarded to syslog/SIEM.	
SR/CR 2.9	Audit storage capacity	Confirm the local log is sized/rotated and offloaded to remote syslog/SIEM, with alerting if the syslog feed fails so events are not lost. **Ext. tools:** Web (log size/remote syslog); SIEM source-silent alert.	Logs sized/offloaded; audit-failure alerting defined.	
SR/CR 2.10	Response to audit processing failures	Confirm a defined response to audit-processing/forwarding failures (alerting; no silent loss). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.9** above (shared steps); confirm this requirement individually and adjudicate it separately.	Response to audit processing failures verified individually.	
SR/CR 2.11	Timestamps	Confirm authenticated **NTP** (or SNTP) time sync so log/syslog timestamps are accurate and correlate. **Ext. tools:** Web (NTP config); Wireshark `ntp`.	Accurate, synchronized timestamps.	
SR/CR 2.12	Non-repudiation	Confirm actions are attributable to a specific user (non-repudiation). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.8** above (shared steps); confirm this requirement individually and adjudicate it separately.	Non-repudiation verified individually.	
NDR 2.13	Physical & diagnostic interfaces	Control physical interfaces: locked cabinet; secure the console and reset; shut/disable unused ports; secure the DI/DO relay wiring; disable unused management services. **Ext. tools:** Physical inspection; web/CLI (unused ports shut; services off).	Console/reset/unused-ports controlled; enclosure sealed.	NDR-specific.

FR 3 — System Integrity (SI)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 3.1	Communication integrity	Management via **HTTPS/SSH/SNMPv3** only (disable HTTP/Telnet/SNMPv1-2c); data-plane segmentation via VLAN/ACL; confirm no cleartext management on the wire. **Ext. tools:** Wireshark (encrypted mgmt); `nmap -p 23,80,161 <sw>` (cleartext off).	All management encrypted; cleartext protocols disabled.	
SR/NDR 3.2	Protection from malicious code	Confirm **secure boot** + **signed firmware** (62443-4-2 line): the switch runs only Moxa-signed firmware and rejects unsigned/altered images; verify the running firmware integrity/status. **Ext. tools:** Web/CLI (firmware/secure-boot status); Moxa hash cross-check; attempt an unsigned image (isolated) -> rejected.	Secure boot + signed firmware; only signed firmware runs.	NDR-specific.
SR/CR 3.3	Security functionality verification	After any change re-verify: HTTPS/SSH/SNMPv3, HTTP/Telnet off, 802.1X, ACL/DHCP-snooping/storm-control, login lockout, and firmware, against the Moxa hardening-guide baseline; back up + compare the config. **Ext. tools:** Config backup/compare; `nmap -p 22,23,80,443,161 <sw>` (hardened state).	Config matches the hardening baseline; unauthorized change detected.	
SR/CR 3.4	Software & information integrity	Confirm software & information (config/program) integrity; unauthorized change prevented/detected. **Steps:** for this product this is assessed with the same evidence as **SR/CR 3.3** above (shared steps); confirm this requirement individually and adjudicate it separately.	Software & information integrity verified individually.	
SR/CR 3.5	Input validation	Confirm the switch tolerates malformed management/protocol frames without instability (hardened stacks + storm control). **ISOLATED only:** fuzz on a bench switch. **Ext. tools:** Protocol fuzzer (ISOLATED); web/CLI CPU/stability; storm-control counters.	No instability on malformed input.	Do NOT fuzz an in-service switch.
SR/CR 3.6	Deterministic output	**N/A.** A switch does not drive a physical process output. Confirm the ring/link failure behavior (Turbo Ring/Chain, RSTP/MSTP convergence) is deliberate. **Ext. tools:** Web (Turbo Ring/RSTP config); ring-break test (lab) -> heal within spec.	N/A - justification recorded; ring/RSTP behavior verified.	N/A
SR/CR 3.7	Error handling	Confirm graceful error handling without info leak; review the event log; err-disable/port behavior deliberate. **Ext. tools:** Web/CLI (event log / port status).	Graceful error handling; no info leak.	
SR/CR 3.8	Session integrity	HTTPS/SSH/SNMPv3 sessions are integrity-protected against replay/hijack; disable weak SSH/TLS ciphers. **Ext. tools:** `ssh-audit <sw>`; `sslscan <sw>:443`.	Strong session crypto; weak algorithms disabled.	
SR/CR 3.9	Protection of audit information	Protect the syslog transport (syslog-over-TLS where supported) and the SIEM store; restrict who can clear the local log (admin-only); protect config backups. **Ext. tools:** Web (log-access control); SIEM access review; syslog transport.	Audit records protected from unauthorized read/modify/delete.	
NDR 3.10	Support for updates	Update firmware only with Moxa-signed firmware via the web UI/CLI; the switch validates the signature (secure boot); plan a maintenance window with rollback to the prior signed version. **Ext. tools:** Web/CLI (signed update/verify); Moxa hash/advisory.	Signed updates with tested rollback; authenticity enforced.	NDR-specific.
NDR 3.11	Physical tamper resistance/detection	Locked cabinet; control console/reset; the event log + syslog record config/firmware changes and restarts as tamper evidence; seal where required. **Ext. tools:** Physical inspection (lock/seal); event-log/syslog review.	Cabinet locked/sealed; interfaces controlled; changes logged.	NDR-specific.
NDR 3.12	Provisioning product-supplier roots of trust	Secure boot anchors the Moxa supplier root of trust: the switch runs only Moxa-signed firmware and rejects unsigned/altered images. **Ext. tools:** Web/CLI (secure-boot/firmware status); attempt an unsigned image (isolated) -> rejected.	Supplier root of trust (secure boot) intact; unsigned images rejected.	NDR-specific.
NDR 3.13	Provisioning asset-owner roots of trust	Confirm the asset owner can provision its own certificates for HTTPS/802.1X (import CA/cert; manage the trust store); default self-signed replaced. **Ext. tools:** Web (certificate import / trust store); confirm enterprise CA used.	Asset-owner certificate provisioning verified; defaults replaced.	NDR-specific.
NDR 3.14	Integrity of the boot process	Confirm secure boot validates firmware integrity before execution so only signed firmware boots; capture the firmware/secure-boot status. **Ext. tools:** Web/CLI (secure-boot status + firmware version).	Secure boot verified; only signed firmware boots.	NDR-specific.

FR 4 — Data Confidentiality (DC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 4.1	Information confidentiality	Confirm information confidentiality — sensitive data is protected in transit and at rest. **Steps:** for this product this is assessed with the same evidence as **SR/CR 3.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Information confidentiality verified individually.	
SR/CR 4.2	Information persistence	On decommission/RMA, reset to factory defaults (clears config, accounts, keys, certs) and remove saved configs/backups so nothing sensitive is recoverable. **Ext. tools:** Web/CLI (factory reset); confirm saved configs removed.	Config, credentials, keys, and certs purged on decommission.	
SR/CR 4.3	Use of cryptography	Confirm modern crypto: TLS 1.2/1.3 (HTTPS), strong SSH ciphers, SNMPv3 (SHA/AES); CA-signed certs; disable weak/legacy protocols (HTTP/Telnet/SNMPv1-2c). **Ext. tools:** `testssl.sh https://<sw>`; `ssh-audit`; SNMPv3 config.	Modern TLS/SSH/SNMPv3; weak protocols disabled.	

FR 5 — Restricted Data Flow (RDF)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 5.1	Network segmentation	Use **802.1Q VLANs** + **ACLs** to isolate zones and a dedicated management VLAN; unused ports in an unused VLAN + shut; confirm the VLAN/ACL/port map matches the zone model. **Ext. tools:** Web (VLAN/ACL/port map); `nmap` across VLANs to prove isolation.	Effective VLAN/ACL segmentation; management VLAN isolated; unused ports shut.	
SR/NDR 5.2	Zone boundary protection	ACLs at the switch + the connected firewall enforce the boundary; management-access control restricts the mgmt plane; only required flows cross. **Ext. tools:** `nmap` at the boundary; ACL/firewall review.	Boundary enforces least-flow; management plane restricted.	
SR/NDR 5.3	General-purpose person-to-person comms restrictions	**N/A.** Network switch with no general-purpose person-to-person messaging. **Ext. tools:** Confirm no such services.	N/A - justification recorded.	N/A
SR 5.4	Application Partitioning	**N/A.** Single-purpose device; no separable applications to partition (handled at the host/SCADA layer).	N/A - justification recorded (not assessed as a control).	N/A

FR 6 — Timely Response to Events (TRE)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 6.1	Audit log accessibility	Confirm the event log + remote **syslog** are readable/exportable and forwarded to the SIEM; **MXview/MXsecurity** + a passive OT IDS provide continuous monitoring. **Ext. tools:** Web log export; syslog + SIEM; MXview/MXsecurity; passive OT IDS.	Events accessible/exportable and monitored in near-real-time.	
SR/CR 6.2	Continuous monitoring	Confirm continuous monitoring — events forwarded to and monitored by a SIEM/OT-IDS in near-real-time. **Steps:** for this product this is assessed with the same evidence as **SR/CR 6.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Continuous monitoring verified individually.	

FR 7 — Resource Availability (RA)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 7.1	Denial-of-service protection	Confirm resilience: **storm control**, **DHCP snooping**, rate limiting, MAC protections, and **Turbo Ring/Chain** + RSTP/MSTP redundancy bound floods and heal fast; management-access control limits the mgmt-plane surface. **Ext. tools:** Web (storm-control/DHCP-snooping/Turbo-Ring status); ISOLATED flood/ring-break test.	Storm control + ring redundancy keep the switch forwarding under fault/load.	
SR/CR 7.2	Resource management	Confirm bounded resources (MAC table, connection/session limits, CPU protection) so a flood cannot exhaust the switch. **Ext. tools:** Web/CLI (resource/CPU status); bench load test.	Resources bounded; no exhaustion-based loss of forwarding.	
SR/CR 7.3	Control system backup	Confirm the config is backed up (versioned, offline; MXview/MXconfig where used) and a restore is tested on a spare switch for a known-good rebuild. **Ext. tools:** Config backup; restore test (spare switch).	Config backup current; restore/rebuild tested.	
SR/CR 7.4	Control system recovery & reconstitution	Confirm recovery/reconstitution — a restore to a known-good state is tested. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.3** above (shared steps); confirm this requirement individually and adjudicate it separately.	Control system recovery & reconstitution verified individually.	
SR 7.5	Emergency Power	**N/A.** Emergency/backup power is provided at the facility/UPS level, not the device; confirm the device retains its security state across a power transfer.	N/A - justification recorded (not assessed as a control).	N/A
SR/CR 7.6	Network & security configuration settings	Harden per the Moxa Security Hardening Guide: disable HTTP/Telnet/SNMPv1-2c and unused services, change defaults, enable login lockout + password policy, restrict management access, enable 802.1X/ACL/DHCP-snooping/storm-control, and patch firmware. **Ext. tools:** Web (services/security config); `nmap -sTU -p 23,80,161,443,22 <sw>` (only required reachable).	Least functionality; insecure services disabled; hardened per the guide.	
SR/CR 7.7	Least functionality	Confirm least functionality — unused services/ports/protocols are disabled. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Least functionality verified individually.	
SR/CR 7.8	Component inventory	Confirm the switch (model, firmware, ports/modules) is inventoried so vulnerabilities track against Moxa PSIRT / CISA advisories; MXsecurity aids inventory. **Ext. tools:** `nmap -sU -p 161 --script snmp-info`; MXview/MXsecurity inventory; Moxa PSIRT feed.	Accurate model/firmware inventory maintained; advisories tracked.	

External Test Tools — how to use

Tool / method	Verifies	How to use — steps / example	Constraints
Web UI (HTTPS) / CLI (SSH)	Native config review (all FRs)	Log in over HTTPS/SSH (change default admin/moxa); review Account Management (users/policy/lockout), Security (802.1X/ACL/DHCP-snooping), firmware/secure-boot status, VLAN, Turbo Ring; back up the config.	Read/authorized; unlock only for change tests.
MXview / MXsecurity	NMS monitoring, inventory & hardening (2.8, 6.x, 7.8)	Centralized monitoring, config/version inventory, and security posture across the Moxa fleet.	Read-only where possible.
nmap	Port/service exposure & segmentation (1.x, 3.1, 5.x, 7.6)	`nmap -sTU -p 22,23,80,443,161 <sw>`; `nmap -sU -p 161 --script snmp-info`; scan per VLAN; confirm 23/80 closed.	Read-only; authorized.
sslscan / testssl.sh	HTTPS TLS strength (1.8, 3.8, 4.3)	`sslscan <sw>:443` / `testssl.sh https://<sw>`; TLS1.2/1.3 only, strong ciphers, valid cert.	Read-only/safe.
ssh-audit	SSH algorithm strength (1.5, 1.8, 3.8, 4.3)	`ssh-audit <sw>`; strong KEX/cipher/MAC; weak algos disabled.	Read-only.
Wireshark / tcpdump	Comm integrity & monitoring (1.2, 3.1, 6.2)	SPAN/tap; confirm mgmt encrypted, EAPOL on 802.1X ports; inspect Turbo Ring/RSTP and NTP.	Passive; SPAN/tap only.

Tool / method	Verifies	How to use — steps / example	Constraints
802.1X supplicant / RADIUS	Port authentication (1.2)	Present authorized then unauthorized endpoints on an 802.1X port; confirm grant + rogue quarantine in RADIUS logs.	Coordinate with NAC team.
Hydra	Login lockout verification (1.11)	Authorized repeated HTTPS/SSH logins to confirm the lockout policy engages.	ISOLATED/authorized only.
Syslog/SIEM + passive OT IDS	Audit & monitoring (2.8, 6.x)	Confirm the switch forwards security syslog; passive OT IDS for anomaly detection.	Coordinate with SOC.
Moxa PSIRT / CISA advisories	Vuln tracking (3.2, 3.10, 7.8)	Track EDS/MDS advisories and required firmware.	Reference.

Controlled - Project Use · Independent Assurance Services · ISA/IEC 62443-3-3 / 4-2 · Product-specific companion to the 62443 assessment tool; confirm exact menu paths / CLI against the deployed firmware and the project hardening baseline. ISOLATED tools run only in a representative/lab environment with written authorization.